Leveraging Quantum Computing and Optimization to Estimate Financial Crashes in Small and Medium-Sized Enterprises

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***Abstract*—In this study, we focus on estimating financial crashes within a network of small and medium enterprises (SMEs) that are customers of Yapi Kredi Bank. These SMEs have complex financial relationships involving receivables and payables among each other. We developed a quadratically constrained quadratic program as a mathematical optimization model aiming at identifying significant vulnerabilities and potential failure points in this network. By defining key parameters such as crossholding ratios, self-holding fractions, asset ownership proportions, and failure cost functions, we capture the intricate dependencies between these entities. The model is inherently nonlinear due to the quadratic terms and constraints, reflecting the real-world complexities of financial networks. Solving this NP-hard problem using classical methods is challenging; therefore, we leverage quantum computing technology to efficiently explore the solution space and produce decision variables that indicate the likelihood of financial distress for each SME in the network. The difficulty of this problem lies in the potential domino effect, where a single delayed payment can trigger subsequent financial distress among connected entities, leading to a cascading collapse within the financial network. Classical computing solutions for such problems would take an impractical amount of time, estimated at 13.7 billion years. In contrast, using quantum computing, our optimization model solves the problem in approximately 7 seconds. Using financial data from 2019, our model successfully identified 90.04 million TL out of 107 million TL of non-performing loans. Similarly, using data from 2020, it identified 669.92 million TL out of 726 million TL of non-performing loans. Overall, the model accurately predicted 91.2.**

Keywords*—Quantum computing, Financial crash estimation, Quadratic Unconstrained Binary Optimization (QUBO), Financial network, Small and Medium-Sized Enterprises (SMEs), Risk analysis, Financial contagion, Cascade effect, Financial system equilibrium

I. Introduction

Financial crash estimation problem has been a significant subject within the finance sector, focused on the prediction of prospective collapses in capital markets or organizations. These financial c rashes are likely to have devastating impacts, resulting in crucial economic repercussions and diminished investor confidence. Accurately and timely estimating financial crashes are essential for preserving economic stability, achieving risk mitigation, and making informed decisions to safeguard investments and avoid extensive financial crises.

Conventional methods of financial crash estimation usually involve complex mathematical models and historical data analysis. However, these methods face serious challenges in dealing with the high-dimensional, non-linear, and interdependent characteristics of financial networks. On the contrary, quantum computing offers a revolutionary approach. Utilizing the principles of quantum mechanics, quantum computing is able to process large amounts of data simultaneously and solve complex optimization problems which is not feasible to solve using classical computers in a reasonable time frame.

Utilization of quantum computing for financial crash estimation problem not only enhances computational efficiency but also offers new insights to analyze complex financial systems. This interdisciplinary approach has the potential to significantly improve the robustness and reliability of crash predictions, offering valuable insights for financial analysts, regulators, and policymakers. Quantum algorithms, specifically designed to handle challenges in optimization models, are used to predict more precise and efficient financial crashes compared to traditional methods. Through this research, we aim to showcase the applicability of quantum computing in addressing one of the most challenging problems in finance. In this study, we propose a pioneering approach for financial crash estimation problem by leveraging quantum computing techniques. Our methodology focuses on formulating a quadratically constrained optimization model that considers the intricate dependencies and dynamics of financial networks.

The rest of this paper is organized as follows: Section 2 reviews the related work and current methodologies in financial crash estimation and quantum computing applications in finance. Section 3 describes the problem analyzed in this study. Section 4 outlines our approach, detailing the proposed quadratically constrained optimization model. Section 5 provides the computational analysis and results. Finally, Section 6 presents concluding remarks and discusses potential directions for future research.

II. Related Work

The field of financial crash estimation has seen remarkable advancements, particularly with the integration of quantum

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computing and network theory. In recent years, several studies have explored different approaches to understanding and predicting financial instability, especially within networks of small and medium-sized enterprises (SMEs).

Elliott, Golub and Jackson (2014) conducted a seminal study on cascades of failures within financial networks, focusing on how discontinuous changes in asset values trigger further failures and how these effects depend on network structure [1]. Their findings highlighted the complex interplay between integration (greater dependence on counterparties) and diversification (more counterparties per organization). They demonstrated that while diversification initially allows cascades to travel, it eventually provides better insurance against failures. Integration also presents a trade-off, balancing increased dependence on other organizations with reduced sensitivity to individual investments. This study illustrated these dynamics with data on European debt cross-holdings, providing crucial insights into financial contagion and network robustness.

Building on the theme of network sensitivity, Hemenway and Khanna (2016) addressed the computational complexity inherent in financial networks [2]. They extended the model proposed by Elliott, Golub, and Jackson to show that small changes in investments by individual institutions can significantly impact the entire network's stability, particularly when institutions have minimal self-holdings. They also demonstrated that even with complete information, estimating the number of potential failures following a shock is computationally intractable. This work underscored the difficulties regulators face in predicting systemic risks and the need for precise information and robust analytical tools. With this they have actually shown this financial crash estimation problem being an NP-Hard problem.

Quantum computing emerged as a promising tool for tackling these challenges, as highlighted in several recent studies. Orus, Mugel and Lizaso (2019) demonstrated that predicting financial crashes, an NP-hard problem, could be approached using quantum annealers [3]. They mapped the equilibrium condition of a simplified financial network to a spin-1/2 quantum Hamiltonian problem, showing how adiabatic quantum computation could assess financial equilibrium more efficiently. This approach provided a novel method for forecasting financial crashes, potentially offering regulators a powerful new tool for risk assessment.

Further exploring quantum applications, Orus, Mugel and Lizaso (2019) reviewed various quantum optimization algorithms and their potential uses in finance [4]. They discussed how quantum annealers could optimize portfolios, identify arbitrage opportunities, and perform credit scoring. Additionally, they highlighted the potential for quantum machine learning to enhance financial deep learning methods and discussed quantum amplitude estimation's applications in Monte Carlo sampling for derivative pricing and risk analysis. This comprehensive review illustrated the broad potential of quantum computing to revolutionize financial problem-solving.

In a similar fashion, Orus, Mugel and Lizaso (2020) examined specific use cases of quantum optimization in finance, particularly focusing on financial crash prediction and dynamic portfolio optimization [5]. They compared different quantum strategies, including quantum annealers and quantum-inspired tensor networks, showcasing their potential to solve complex financial problems. Their analysis reinforced the idea that quantum computing could significantly enhance the efficiency and accuracy of financial modeling and optimization. They also mention predicting financial crash estimation with a cross-holdings of 20 financial institutions would take more than 13.7 billion years to observe the effect of perturbation in the network.

More recently, Ding et al. (2021) experimentally explored the prediction of financial crashes using a D-Wave quantum computer [6]. They embedded a nonlinear financial model's equilibrium condition into a higher-order unconstrained binary optimization problem, which was then transformed into a spin-1/2 Hamiltonian. This approach aimed to approximate financial equilibrium via quantum annealing, paving the way for practical applications of quantum computers in macroeconomic modeling. Despite current hardware limitations, their work demonstrated the feasibility of using quantum computing for complex financial predictions.

Lastly, Herman et al. (2022) provided a comprehensive survey of quantum computing applications in finance [7]. They reviewed the state of the art in quantum algorithms for Monte Carlo integration, optimization, and machine learning, emphasizing their potential to solve problems such as derivative pricing, risk analysis, portfolio optimization, natural language processing, and fraud detection. This survey underscored the transformative impact of quantum computing on finance, highlighting its potential to improve efficiency and accuracy across various financial applications.

III. Problem Description

In this study, financial crash estimation problem is formulated within the context of a financial network comprising small and medium-sized enterprises (SME) of Yapı Kredi Bank. Identifying how small disruptions in the financial network could lead to a total financial crash is a key to gain insights regarding the risk of lending and investment in the SME customers. The motivation behind this study is to comprehend the risk of financial collapse over the network when the organizations in the network, i.e SME customers of the bank, are financially interdependent.

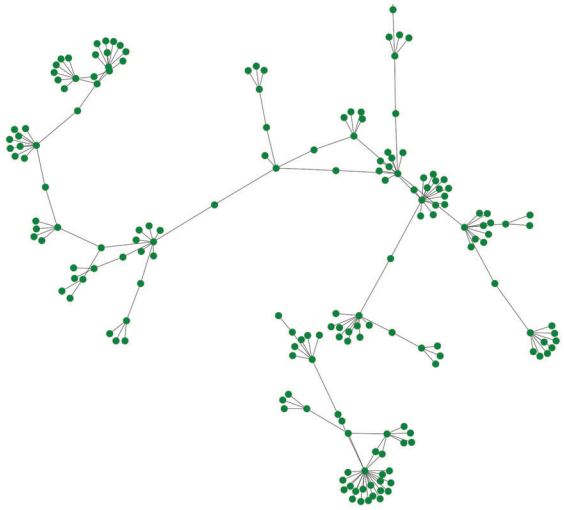

Fig. 1. Example of a financial network.

A. Setting

Constructing the accurate financial network of SMEs is a key to address the financial crash estimation problem. In this section, an overview of the financial network that we analyze in this study is provided, focusing on the financial dependencies and relationship among the entities within the network. The

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example of a financial network depicted in Fig. 3 demonstrates a chosen collection of small and medium enterprises (SMEs) being the customers of Yapi Kredi Bank. Each node in the network represents a singular SME, while the edges connecting the nodes correspond financial transactions such as accounts receivable or account payable. Integrating these financial transactions into the network brings about a bundle of interdependencies which could remarkably impact the financial equilibrium of the entire network.

The paper Ding et. al published has also investigated the problem of financial crash estimation with D-wave using their quantum annealers similar to what we were investigating. However, there were major differences compared in their setting. Initially, their data consisted direct cross-holdings between the firms where in our case we had a debt relation between the institutions where we interpreted these as crossholding as explained the section IV. Another major difference is that they were approaching the problem as a QUBO (quadratic unconstrained binary optimization) problem whereas we benefited from dwave's hybrid solvers to model our problem with a CQM (constrained quadratic model). [6]

A general perspective in the literature of the financial crash estimation depends on the crossholding ratio of firms as well as their market values [3], [4]. However, the institutions that we are working on doesn't have any market values as they are not open to markets and therefore they do not hold any shares of other firms in the network as there aren't any shares for these companies. Hence, this changes the way we approach to the problem. In order to be able to compute financial crashes among such institutions we needed to come up with a new way that would capture the idea behind the financial crash optimization formulation. To provide this we needed to manipulate SMEs financial information which is explained in the later sections.

B. Preliminaries

The selection of SMEs to be represented as nodes in the network was carried out by analyzing financial records of the customers in our database. In this analysis, an extensive review of trial balances and balance sheets of the customers whose Strategic Business Unit (SBU) segment remained unchanged between January 1, 2019 and December 31, 2020 was conducted. By taking into account the customers complying with this characteristics, we determined three distinct types of customers, as shown in Table V.

TABLE I
CUSTOMER TYPES BASED ON FINANCIAL STATUS

Customer Type	Description
Type 1 Customer	Customers who fell into NPL status at least once during the 24-month period
Type 2 Customer	Customers who did not fall into NPL status but were on the WL for at least 3 consecutive months during the 24-month period
Type 3 Customer	Customers who did not fall into NPL status, were on the WL for at most 2 consecutive months, or remained Healthy throughout the 24-month period

In this setting, NPL (Non-Performing Loans) denotes the customers whose payment delays exceed 90 days, whereas WL (Watchlist) refers to the customers with payment delays between 15 and 90 days, and Healthy pertains to customers whose payment delays are less than 15 days. By classifying customers in this fashion, we make sure that our network model precisely reflects the diverse risk profiles and financial behaviors that are present within the SME segment. This classification enable us to make the connection between customer risk profile and potential points of failure and thereby understanding the cascade effects of financial distress within the network. The distribution of these classes of customers in the input dataset can be examined in Figure 2.

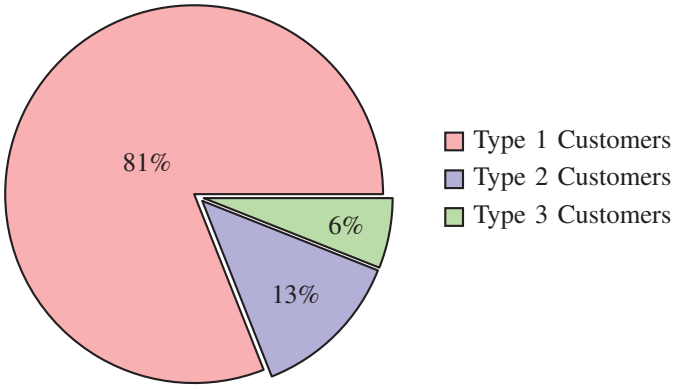

Fig. 2. Distribution of Customer Types in the Financial Network

Financial obligations between each pair of SMEs are illustrated as the edges in the network. For example, if one customer owes money to another, an edge connecting these two customer is drawn from the debtor to the creditor. The weight of the edge corresponds to the debt amount for the debtor, which is the same as the credit amount for the creditor. These connections are significant in forthoming the potential cascade effects leading up to financial crash. When an SME faces with financial difficulty, it might result in delay or default on its payments that triggers a domino reaction influencing the remaining connected SMEs. These interdepencies are incorporated by our optimization model, which enables us to identify critical nodes whose financial failure could propagate financial distress through the entire network. To ensure greater stability within the financial network, we aim to forecast and mitigate the impact of financial crashes by investigating this network structure.

IV. PROPOSED METHOD

According to the work of Orus, Mugel and Lizaso (2022), they stated that the financial crash estimation is a good case for quantum computing because of its complexity level. Dwave's Hybrid solvers has 3 different categories: BQM (Binary Quadratic Model), DQM (Discrete Quadratic Model) and CQM (Constrained Quadratic Model). Since the nature of the problem has constraints, high complexity and integers values the best fit is CQM.Therefore we performed an optimization modeling with CQM through Dwave's hybrid solvers to solve the problem mentioned in Chapter 3. A detailed explanation of the model is below.

A. Parameters

In this section, we describe the parameters used in our mathematical model. These parameters contain coefficients, constants, and other predefined values which influence the formulation of the decision variables and the overall constraints of the model.

To represent the financial relations among the institutions involved in the network, we define the following parameters:

- $C_{ij} \rightarrow$ the fraction of organization j owned by organization i

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TABLE II
DATA DICTIONARY USED IN THE FORMULATION OF THE PARAMETERS

Tableau	Column	Description
Trial Balance	Date	Date of entry for trial balances and balance sheets
	Company A	Company owning the trial balance
	Company B	Company in financial relation with Company A
	Financial Dependency	Indicates if it is a payable or receivable
	Amount debit	Amount owed by Company A
	Amount credit	Amount owed to Company A
Balance Sheet	Cash & Cash Equivalents	Cash, checks, banks, payment orders, other equivalents
	Securities	Stocks, bonds, notes (private and public)
	Trade Receivables	Customers, receivables, discounts, deposits
	Financial Liabilities	Bank loans, bond payments, interest
	Trade Payables	Suppliers, notes payable, deposits
	Advances Received	Order advances, other advances
	Total Equity	Paid-in capital, reserves, retained earnings

- $\hat{C}_{ii} \rightarrow$ the fraction of organization i not owned by any other organization

In other words, the magnitude of the C_{ij} value directly corresponds to the financial dependency between the organization pair (i, j).

The notation and detailed description of each parameter is provided in Table III. In Table III, we explain how we calculate the values of the parameters based on the mathematical formulations we developed for our use case.

B. Decision Variables

The decision variables in our model are described as follows:

- y_i is a binary variable defined as:

$$y_i = \begin{cases} 1, & \text{if } v_i < v_i^C \quad \text{for } i \in I \\ 0, & \text{otherwise} \end{cases}$$

- v_i represents the market value of institution i, where $i \in I$.

C. Mathematical Model

The mathematical model is formulated to achieve the *system equilibrium* by minimizing the total disparity over the financial network. This involves the minimization of the difference between the market value of the institutions and derived market values from the multiplication of $\hat{C}_{ii}$ and V_i, considering the financial dependencies of the institutions among each other through the realization of cross-holdings involved in the equity values.

Let us denote $F(v)$ being the objective function value of the mathematical model as follows:

$$F(v) = \sum_{i \in I} \left(v_i - \hat{C}_{ii} \left(\sum_{j \neq i} C_{ij} V_j + \sum_k D_{ik} p_k - \beta_i(\mathbf{p}) y_i \right) \right)$$

Our model is based on the principle that $F(v) = 0$ when the network is in the financial equilibrium. The system equilibrium function $F(v)$ is defined by Elliott, Golub and Jackson (2014) to forecast financial crashes [1].

We propose the following constrained quadratic modeling approach to estimate financial crashes in our network.

$$\begin{aligned} \min_{\mathbf{v},\mathbf{y}} \quad & \sum_{i \in I} \left(v_i - \hat{C}_{ii} \left(\sum_{j \neq i} C_{ij} V_j + \sum_k D_{ik} p_k - \beta_i(\mathbf{p}) y_i \right) \right)^2 \\ \text{s.t.} \quad & v_i y_i \leq (v_i^C - 1) y_i, && \forall i \in I \\ & v_i^C (1 - y_i) \leq v_i (1 - y_i), && \forall i \in I \\ & y_i \in \{0, 1\}, && \forall i \in I \\ & v_i \in \{0, 1, 2, \ldots, v_i^{\max}\}, && \forall i \in I \end{aligned}$$

The objective function minimizes the total financial imbalance caused by financial failures of the institutions. Constraint 1 ensures that the indicator function y_i denoting the financial failure of the institution is equal to 1 when market value v_i of the institution falls below the given critical threshold value v_i^C. Constraint 2 enforces indicator function y_i being equal to 0 when market value v_i of the institution is greater than or equal to the given critical threshold value v_i^C. Constraint 3 is the domain restriction for the binary variable y_i representing the financial failure of the institutions. Constraint 4 makes sure that the decision variable v_i is restricted to take only integer values from a predefined discrete set.

V. COMPUTATIONAL ANALYSIS

To achieve results from the optimization model that we have constructed, we used Dwave's hybrid solvers allowing classical algorithms to be optimized with Quantum Annealing principle for large problems. With this processing power, we were able acquire a solution to the current optimization problem within 7 seconds. The hybrid solvers approximate the probable expected solution time with the given complex data and the optimization formulation that we want this data to be solved. The proposed time of these hybrid solvers for our problem was 7 seconds. When we wanted to solve the problem in a smaller amount of time the results we were getting were not feasible, meaning that there were constraint violations in the found solution. We have also tried raising the solution time but we have observed that going above the recommended solve time didn't affect the results that we were getting are all feasible solutions. However, due to randomness of the Quantum Processing Mechanism and having multiple equilibrium points in the optimization problem, each different setting has been repeated for 5 times as this allowed solutions to be in possible different optimal equilibrium points in solution space. According to the work of Orus, Mugel and Lizaso [3], they stated that even with small networks (20-30 institutions), solving these type of problem takes 13.7 billion years with classical methods, yet with the mentioned Quantum Processing power we were able to solve the problem in a reasonable amount of time as 7 seconds per solution.

Before starting experimental runs, there are couple of important preprocessing steps to be done in order for the problem to work in QPUs. Initially, the floating numbers should be eliminated since the Quantum Solvers that we had can work with integers only. Therefore, instead of rounding the numbers all numeric data had been increased by 10 to the power of maximum decimal point number that's in the data. Even though

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TABLE III
PARAMETERS DEFINED FOR THE QUADRATICALLY CONSTRAINED OPTIMIZATION MODEL

Parameter	Symbol	Description
Crossholding Ratio	C_{ij}	The ratio of the debt that organization i owes to organization j, relative to the overall debt that organizations collectively owe to organization j
Self-Holding Fraction	$\hat{C}_{ii}$	The fraction of organization i's ownership that remains unaccounted for by other organizations' debts: $\hat{\mathbf{C}}_{\mathbf{ii}} = [\mathbf{1} - \sum_{\mathbf{i} \neq \mathbf{j}} \mathbf{C}_{\mathbf{ji}}]$
Asset Ownership Proportion	D_{ik}	The proportion of the market price of asset k held by organization i
Market Price	p_k	The overall market price of asset k held by all organizations in the network
Book Value	V_i	The book value (equity) of organization i: $\mathbf{V} = (\mathbf{I} - \mathbf{C})^{-1}\mathbf{Dp}$
Failure Cost Function	$\beta_i(\mathbf{p})$	The failure cost incurred by organization i when its market value v_i falls below a critical value v_i^C: $\beta_{\mathbf{i}}(\mathbf{p}) = \sum_{\mathbf{k}} \lambda_{\mathbf{ik}} \mathbf{D}_{\mathbf{ik}} \mathbf{p}_{\mathbf{k}}$
Threshold Value	v_i^C	The threshold value for market values of each organization i: $\mathbf{v}^{\mathbf{C}} = \theta(\mathbf{A}^{\mathbf{20}}\mathbf{D}^{\mathbf{20}}\mathbf{p}^{\mathbf{20}})$ where $\theta \in \{.9, .93, .935, .94, .945\}$
Dependency Ratio	A_{ij}	The dependency relation representing how vulnerable organization j is to loss in the debt value of organization i (the degree of potential negative impact on j if i fails to pay its debt to j): $\mathbf{A} = \hat{\mathbf{C}}(\mathbf{I} - \mathbf{C})^{-1}$
Unit Liquidation Cost	λ_{ik}	The unit cost incurred to organization i if asset k is liquidated

this has increased the magnitude of numbers, we were able to preserve numerical relation between the customers.

Due to the nature of the problem there exists multiple equilibrium points indicating which firms are going to have a possible bankrupt and which are not if we project these points as in real world banking solutions. If the current financial statements were covering the whole universe of the firms in the problem, then all the equilibrium point observations would be viable solutions to the problem. Since current financial information and relations between the companies are inadequate equilibrium points required to be post-processed. For instance a company may have a large amount of receivable outside of the network that is not observable with the current data, may actually boost the status of the firm in the real world. Yet, according to the available data this firm could be failing in giving circumstances. Under such conditions, using classical machine learning metrics like precision and recall would be meaningless to evaluate the performance of the model.

As we have investigated the equilibrium points and the results of the model, real world unpaid debt information was the best metric to evaluate the model and its parameters. When the equilibrium point is stated from the model, we have checked how much of the real world unpaid debt could be observed from model's indicated crashed firms. The different quarter results are shown in Table IV.

TABLE IV
MODEL PREDICTED UNPAID DEBT

Quarter	Predicted Unpaid Debt M TL	Real Unpaid Debt M TL	Accuracy %
4	90.98	94.81	95.95
8	81.35	87.77	92.69

The reason for considering 4th and 8th quarter from the data is because these two end of the year quarters are the ones reflecting real world financial situations of the companies as they provide more comprehensive financial statements and documents compared to other fiscals of year. Although such results, these conditions were not fully exploring the failing of the firms since we were considering the current financial results of the companies. What we have done next, is we have investigated the status of failing companies through looking at their future financial terms while solving the optimization problem in the current financial data. Since we were working on the past data we could observe the potential success of the model through the next months of the conducted prediction. For instance if the experiment is done on the 2019 quarter 4, we have analysed the unpaid debt of a company in 2020 quarter 1, quarter 2 etc. Therefore we have extended the previous experiment through considering quarter 4 (2019 Q4) and quarter 8 (2020 Q4) results in the future conditions of the firms.

TABLE V
MODEL PREDICTED UNPAID DEBT WITH NEXT QUARTERS

Quarter	Predicted Unpaid Debt M TL	Real Unpaid Debt M TL	Accuracy %
4	90.98	107.0	84.13
8	669.92	726.0	92.28

With the dependency matrix between the firms, A, we could observe how companies' failing according to the model will affect the debt holdings of the other firm. An example is the figure 3 where a company with bad regarded financials, D, is failing according to the model. On the edges, company D's dependency values to K, C, P, T could be observed. Due to the nature of cascading the financial crashes effect could be observed in the next quarter, and on point we could observe that the failing of firm D has affected the company P as it has now 59 thousand TL in unpaid debt, meanwhile D's unpaid Debt has increased to 167 thousand.

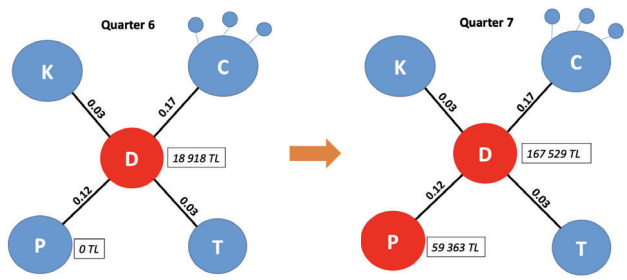

Fig. 3. Example of a Cascade Effect

VI. CONCLUDING REMARKS

As remarked in the computational analysis part, optimization problem is required to be repeated multiple times for achieving different equilibrium points. In our case even some equilibrium points are viable in mathematical sense, in banking terms they were not feasible. In future works it would be more practical to develop a testing mechanism to achieve applicable results from

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the solution set while decreasing the mentioned repetition of 5 times.

If banks can ensure more frequent and complete collection of companies' trial balances, the quality of the networks created can directly improve the quality of the work. Likewise, if large companies can be included in the network of small-medium-sized companies, a more realistic projection will be provided.

Another future work that can be done is to take statistical modeling one step further by getting help from experts in the field to benefit from different and more diverse documents, not just trial balance documents.

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